Reasoning-Aware Pruning of Large Language Models Without Fine-Tuning

Jai Pavan Ramani (jramani3), Arnav Yennam (aryennam), Shivang Nitinkumar Patel (spatel74)

Problem Statement

Large Language Models (LLMs) are highly over parameterized, making pruning an attractive approach for reducing inference cost. However, existing pruning methods rely on magnitude or activation based importance metrics that optimize for average language modeling rather than specific capabilities. As a result, when applied without retraining, these methods often preserve perplexity but significantly degrade multistep reasoning tasks such as arithmetic and logical inference. Post pruning fine tuning can recover performance but it is computationally expensive for large models.

This project aims to address the problem by designing a pruning strategy that preserves reasoning capabilities in pretrained LLMs without any fine-tuning. The goal is to identify layers or parameters whose removal disproportionately harms reasoning accuracy, develop a metric that captures reasoning sensitivity, and incorporate it into a oneshot, capability aware pruning framework that selectively protects reasoning critical components.

Motivation

Our motivation behind this idea was simply the fact that pruning is very well explored in classification models, but less explored in LLM Reasoning models. This is because current pruning approaches for LLMs degrade performance on multi-step reasoning tasks such as arithmetic and logical inference, and the retraining step after pruning is very expensive for really large models. Our goal is to identify the layers or parameters whose removal really harms reasoning performance by developing some sort of metric to measure the same. We then intend to use the metric to identify and protect reasoning critical components while pruning others, with the hope that this will maintain proper reasoning performance even without retraining.

Plans on what to do

- Choose a pretrained LLM and evaluate it on reasoning benchmarks and non-reasoning metrics. Apply an existing pruning method without retraining to demonstrate that reasoning performance degrades significantly.
- Individually disturb or scale the outputs of each transformer layer during inference and measure the resulting drop in reasoning accuracy and language modeling performance. Use this to identify layers that are disproportionately important for multi-step reasoning.
- Construct a pruning metric that combines standard weight/activation importance with layer-level reasoning sensitivity, prioritizing components that are critical for reasoning.

- Prune the model in a one-shot manner while protecting reasoning-critical layers or parameters, without any fine-tuning or retraining.
- Compare the pruned model against standard pruning baselines on reasoning and non-reasoning tasks to assess how well reasoning performance is preserved.

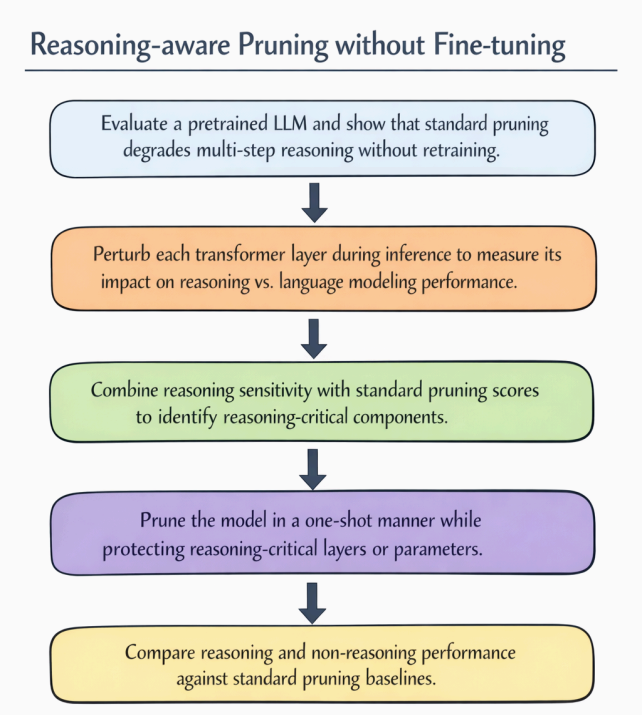


Figure 1: Flowchart showing our planned steps in brief.

Expected Results

We expect standard pruning methods to cause disproportionately large drops in reasoning accuracy compared to general language modeling metrics, even at moderate sparsity levels. While perplexity or general task performance may degrade gradually, reasoning benchmarks such as arithmetic or logical inference are likely to collapse much earlier, highlighting the mismatch between average importance metrics and structured reasoning capability.

Layer wise disturbance experiments are expected to show that reasoning sensitivity is not uniformly distributed across the network. Certain transformer layers, likely mid to late layers, should exhibit significantly larger drops in reasoning accuracy when perturbed, indicating that reasoning-related computations are partially localized. In contrast, some layers may affect general language modeling more than reasoning, reinforcing the idea that different capabilities rely on different internal structures.

Using a pruning metric that combines traditional importance measures with reasoning sensitivity, the proposed method should better distinguish reasoning-critical components from non-critical ones. At equivalent sparsity levels, the reasoning

aware pruned model is expected to retain substantially higher reasoning accuracy than baseline pruning approaches, while maintaining comparable perplexity. Although there may be a slight trade-off in maximum achievable sparsity, the degradation in reasoning performance should be significantly more gradual and controlled, demonstrating that reasoning pathways can be preserved without any retraining.

Literature Survey

Wang et. al [1] proposed a self-reflective pruning framework which uses model's own reasoning traces to estimate parameter importance, allowing pruned models to better preserve multi-step reasoning performance than traditional pruning methods.

Liu et. al [2] introduced a hybrid-grained structured pruning method which combines fine and coarse grained importance metrics to improve overall task performance after pruning.

Role of each person (flexible and could change over time)

Jai: He will be working on identifying critical components of the model in terms of selection of reasoning benchmarks and computing reasoning-sensitivity metrics to rank components based on their impact on reasoning accuracy.

Arnav: He will be responsible for designing and implementing the pruning strategy. This involves designing an algorithm which preserves reasoning-critical components while aggressively pruning less important parts.

Shivang: He is responsible for experimental evaluation and final system validation. This includes running baseline and proposed pruning methods and comparing reasoning and general task performance.

References

[1] Xiang, Yang, Yixin Ji, Juntao Li, and Min Zhang. "Think Before You Prune: Selective Self-Generated Calibration for Pruning Large Reasoning Models." *arXiv preprint arXiv:2511.18864* (2025).

[2] Liu, J., Kong, Z., Zhao, P., Yang, C., Shen, X., Tang, H., Yuan, G., Niu, W., Zhang, W., Lin, X., Huang, D., & Wang, Y. (2025). Toward Adaptive Large Language Models Structured Pruning via Hybrid-grained Weight Importance Assessment. *Proceedings of the AAAI Conference on Artificial Intelligence*, *39*(18), 18879-18887. https://doi.org/10.1609/aaai.v39i18.34078